MILITARY STANDARD

Packaging of Petroleum

and Related Products

To All Holders of MIL-STD-290E

1. The following pages of MIL-STD-290E have been revised and supersedes the page listed:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
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iii	1 Aug 1985	iii	1 May 1981
iv	1 Aug 1985	iv	1 Sep 1982
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3. Holders of MIL-STD-290E will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or cancelled.

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MIL-STD-290E
1 May 1981
SUPERSEDING
MIL-STD-290D
8 November 1976

MILITARY STANDARD

PACKAGING OF PETROLEUM

AND RELATED PRODUCTS

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20402

Packaging, Packing and Marking of Petroleum and Related Products

MIL-STD-290

1. This Standard has been approved by the Department of Defense and is mandatoryfor use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Naval Supply Systems Command (SUP 061), Washington, DC 20376, by using the self-addressed Standardization Document Improvement Proposals (DD Form 1426) appearing at the end of this document or by letter.

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MH2.15-1979	5 Gallon (19 Liter) Nesting Lug-Cover, Universal Steel Pails (DOT 37A80, DOT 37A60 UFC-Rule 40 NMFC-Item 260)
MH3.1-1974	- Round Metal Motor Oil Cans
MH3.3-1975	- Grease Cartridge
MH3.4-1976	- Round Composite Motor Oil Cans
Z-129.1-1976	- Precautionary labeling of Hazardous Industrial Chemicals

(Copies may be purchased from the American National Standards Institute, Inc., 1430 Broadway, New York 10018.)

ASTM-D-3951 Standard Practice for Commercial Packaging

(Copies may be purchased from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Freight Classification; Uniform Classification Committee, Agent

(Copies may be purchased from the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. DEFINITIONS

3.1 Commercial packaging. The methods and materials employed by the supplier to satisfy the requirements of the commercial distribution system.

3.2 Exterior pack. A container or bundle which is sufficient by reason of material, design and construction to protect material during shipment and storage. This can be the unit pack or a container with any combination of unit or intermediate packs.

3.3 Intermediate pack. A wrap or box which contains two or more identical unit packs.

3.4 Levels of protection. The degree of preservation or packing required to prevent deterioration or damage to supplies and equipment due to the hazards to which they may be subjected during shipment and storage.

3.5 Marking. Application of numbers, letters, labels, tags, symbols or colors for handling or identification during shipment and storage.

3.6 Shelf-life item. An item of supply possessing deteriorative or unstable characteristics to the degree that a storage time period must be assigned to assure that it will perform satisfactory in service.

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3.6.1 Type I Shelf-life item. An item of supply which is determined through an evaluation of technical test data and/or actual experience, to be an item with a definite nonextendible period of shelf life.

3.6.2 Type II Shelf-life item. An item of supply with an assigned storage time period which may be extended after the completion of prescribed inspection and/or restorative action.

3.7 Unit Pack. The first tie, wrap or container applied to a single item or a quantity thereof, or to a group of items of a single stock number, preserved or unpreserved, which constitutes a complete identifiable pack. For petroleum products, the unti pack is a can, drum, pail, or the pack described in the requirements, Section 4 and 5.

4. GENERAL REQUIREMENTS

4.1 Unit containers. Unit containers for packaging petroleum and related products shall conform to the requirements for level B or C, or commercial as specified in Table I. Closures and seals for unit containers shall be as specified in Table I.

4.2 Coating of metal containers. Metal containers shall be coated (paint type) as specified in Table I and 5.8.

4.3 Packing. Petroleum and related products in unit containers specified in Table I shall be packed level B or C, or commercial as specified in 5.1 through 5.6. (See 6.1) Specialty products shall be packed as specified in 5.7.

4.4 Marking. Packed petroleum and related products will be marked as specified in 5.9.

4.5 Palletization. When specified, packed petroleum and related products will be palletized in accordance with 5.4.1. (See 6.1)

4.6 DOT Regulations. All petroleum and related products regulated by the Department of Transportation shall be packed in containers conforming to 49 CFR. The containers shall also meet the requirements to Air Force Regulation 71-4 when Military Air Lift is involved. Additionally, when specified, petroleum and related products shall be packed in containers meeting the 15 PSI internal air gauge pressure leakage test requirements. (See 5.4 and 6.1)

5. DETAILED REQUIREMENTS

5.1 Unit pack. When petroleum and related products are furnished in composite cans, such cans shall conform to the requirements specified in Table I for the appropriate level as specified in the contract or order. (See 6.1)

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5.1.2 Exterior packing. Filled and closed composite cans shall be packed level B or C, or commercial as specified in the contract or order. (See 6.1)

5.1.2.1 Level B. Filled and closed composite cans, in the quantity and arrangement specified in Table II, shall be packed in V3c, V3s, W5c, or W5s style RSC fiberboard boxes constructed and closed in accordance with PPP-B-636. Boxes shall be top loading or end loading.

5.1.2.2 Level C. Filled and closed composite cans, in the quantity and arrangement specified in Table II, shall be packed in domestic, style RCS fiberboard boxes constructed and closed in accordance with PPP-B-636. Boxes shall be top loading or end loading.

5.1.2.3 Commercial. Filled and closed cans shall be packed in accordance with ASTM D-3951.

5.2 Packing products in unit containers.

5.2.1 Unit pack. When petroleum and related products are furnished in unit containers, such containers shall conform to the requirements specified in Table I for the appropriate level as specified in the contract or order. (See 6.1)

5.2.2 Intermediate packing. Filled and closed containers of less than 1-pint capacity shall be intermediately packed in accordance with ASTM D-3951.

5.2.3 Exterior packing. Filled and closed containers shall be packed level B or C, or commercial as specified in the contract or order. (See 6.1)

5.2.3.1 Level B.

5.2.3.1.1 Intermediate packs of filled containers of less than 1-pint capacity. Intermediate packs in the quantity and arrangement specified in Table II, shall be packed in V3s, V3c, W5s, or W5c style RSC fiberboard boxes constructed and closed in accordance with PPP-B-636. Boxes may be end loading or top loading.

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5.2.3.1.2 Filled containers 1-pint capacity and over. Filled and closed containers of 1-pint capacity and over, in the quantity and arrangement specified in Table II, shall be packed in snug fitting V3s, V3c, W5s, or W5c style RSC fiberboard boxes constructed and closed in accordance with PPP-B-636. Boxes may be end loading or top loading. Containers with protruding closures shall be protected with cell spacers or buffer pads, formed from scored sheets of fiberboard or from polystyrene cushioning materials or full height, snug fitting half-slotted style, interlocking fiberboard partitions. The design number and arrangement of the pads shall prevent movement and provide clearance.

5.2.3.2 Level C.

5.2.3.2.1 Intermediate packs of cans less than 1-pint capacity. Intermediate packs in the quantity and arrangement specified in Table II shall be packed in snug-fitting domestic grade, style RSC fiberboard boxes constructed and closed in accordance with PPP-B-636. Boxes may be top loading or end loading.

5.2.3.2.2. Filled cans 1-pint capacity and over. Filled cans of 1-pint capacity and over shall be packed and cushioned as specified in 5.2.3.1.2 except that the fiberboard boxes shall be domestic grade.

5.2.3.3 Commercial. Filled and closed cans shall be packed in accordance with ASTM D-3951.

5.3. Packing Standard Grease Cartridges (14-ounces nominal contents for Cartridge-type Grease Guns).

5.3.1 Unit packing. Grease shall be furnished in grease cartridges conforming to the requirements in Table 1 for level B or C, or commercial as specified in the contract or order. (See 6.1)

5.3.2 Intermediate packing. Filled cartridges shall be intermediately packed in accordance with commercial practice.

5.3.3 Exterior packing. Exterior packing shall be level B or C, or commercial as specified in the contract or order.

5.3.3.1 Level B. Not more than sixty filled grease cartridges, intermediately packed as specified in 5.3.2, shall be packed in V3c, V3s, W5c, W5s, W6c, or W6s. Boxes may be top loading or end loading.

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5.3.3.2 Level C. Not more than sixty filled grease cartridges, intermediately packed as specified in 5.3.2 shall be packed in domestic grade, style RSC fiberboard boxes constructed and closed in accordance with PPP-B-636.

5.3.3.3 Commercial. Filled cartridges shall be packed in accordance with ASTM D-3951.

5.4 Packing products in drums. When drums are required, drums shall conform to the requirements in Table I for the appropriate level specified in the contract or order. When Military Air Shipment is involved such drums shall also meet the requirements of Air Force Regulation 71-4. (See 6.1)

5.4.1 Palletization. When specified, packed petroleum and related products shall be palletized in accordance with MIL-STD-147. Marking of palletized loads shall be in accordance with 5.9.4.5. Unless otherwise specified, double wing pallets shall be used.

5.4.1.1 Arrangement of Drums on Pallets. When specified, four 55-gallon drums shall be palletized on a 48" X 48" double wing pallet conforming to NN-P-71 as shown in Figure 7. Strapping shall be 3/4" conforming to QQ-S-781. See Figure 7, 55-gallon drum palletization.

5.4.1.2 Unitized Loads for Army Use Only. When specified, three 55-gallon drums shall be palletized on standard 40" X 48" 4 way entry pallets and will be secured to pallets by stretch wrap bounding. In instances when stretch film capacity is unavailable or the commodity is not compatible with the stretch process, drums shall be secured with 3/4" steel strapping conforming to QQ-S-781 applied girthwise. Quantities of less than 3 drums need not be palletized. (See Figure 7a.)

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5.7.2.2.3 Commercial. Wax shall be packed in accordance with ASTM D-3951.

5.7.3 Cleaner, Lubricant And Preservative (CLP).

5.7.3.1 Unit Packing.

5.7.3.1.1 Unit Packing Container Sizes. CLP shall be furnished in one of the following high-density-polyethylene containers:

- 1.1 fl. oz (33 ml) plastic bottle
- 4 fl. oz (120 ml) plastic bottle with extender tube
- 1 gal (3.785 litres) plastic bottle
- 1 pt (480 ml) plastic bottle with trigger sprayer

Commercial packaged CLP shall be adequate for unit packaging requirements.

5.7.3.2 Exterior Packing. CLP shall be packed level A or C as specified. Unit packs per container shall not exceed the following:

- 1.1 fl oz (33 ml) ---------- 240 units/container
- 4 fl oz (120 ml) ---------- 120 units/container
- 1 pt (480 ml) ---------- 20 units/container
- 1 gal (3.785 liters) ---------- 4 units/container

5.7.3.3 Level A. Applicable for overseas shipment only. Shall be packed in wood containers conforming to PPP-B-601, overseas type or PPP-B-621, Class II.

5.7.3.4 Level C. Packing shall be in accordance with ASTM D-3951.

5.8 Exterior Color Coating of Metal Containers.

5.8.1 General. Exterior coatings (paint type) on metal containers shall be TAN, approximating No. 22648 of Federal Standard 595, for all containers except as specified in 5.8.2 through 5.8.5. Certain container areas shall be unpainted as specified in 5.8.2 through 5.8.5. Type I cans requiring unpainted top and bottom lids shall have a 1/4 inch unpainted margin on both ends of 8-ounce can bodies, a 1/2 inch unpainted margin on both ends of 1-quart can bodies an 3/4 unpainted margin on both ends of 1-gallon can bodies. Striping on side seams is not required. Exterior paint is not required on friction rings, on tops and bottoms of 4-ounce capacity or less cans, or on collapsible tubes. The exterior

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tops and bottoms of 5 gallon cans and larger will be painted. When level C or commercial level is specified, exterior painting of metal containers shall conform to either the above specification or the supplier's commercial color, except: (a) that solid RED, YELLOW or GREEN coatings shall be used only as specified in 5.8.2 through 5.8.5. (b) the silver color or uncolored or plain tin-plate cans will not be allowed for the one quart and one gallon type I cans (metal, round, open-top hermetically sealed when filled).

5.8.2 Hydraulic Fluids.

5.8.2.1 Color. For RED or YELLOW colors the enamel shall conform to Specification MIL-E-19603 and the primer to Specification MIL-P-19602. Colors for containers of other hydraulic fluids shall be as specified in 5.8.1. Color number shall conform to those listed in Federal Standard 595.

a. MIL-H-5606. Hydraulic Fluid, Petroleum Base, Aircraft Missile and Ordnance. Containers of one gallon or less and 10-gallon drums shall be coated RED, a color approximating Color No. 11105 or 21105. Except for 10-gallon drums, containers larger than one-gallon shall be coated TAN, approximating Color No. 22648. Exterior tops and bottoms of cans shall not be painted.

b.MIL-H-83282. Hydraulic Fluid, Fire Resistant Synthetic Hydrocarbon Base, Aircraft. Containers of one-gallon or less shall be coated RED, a color approximating Color No. 11105 or 21105; containers larger than one-gallon shall be coated TAN, approximating Color No. 22648. Exterior tops and bottoms of cans shall not be painted.

c. MIL-H-6083. Hydraulic Fluid, Petroleum Base, for Preservation and Operation. Containers of one-gallon or less shall be coated RED, a color approximating Color No. 11105 or 21105; containers larger than one-gallon shall be coated TAN, approximating Color No. 22648. In addition, 5-gallon pails and 55-gallon drums shall have a WHITE band approximately one-inch wide painted on the side directly below the top head chime. Exterior tops and bottoms of cans shall not be painted.

d. MIL-H-46170 Hydraulic Fluid, Rust Inhibited, Fire Retardant, Synthetic Base, Type I and II. Containers of Type I hydraulic fluid shall be coated TAN approximating Color No. 22648. Exterior tops and bottoms of cans shall not be coated. Containers of Type II hydraulic fluid of one-gallon or less shall be coated RED, a color approximating Color No. 11105 or 21105; containers larger than one-gallon shall be coated TAN approximating Color No. 22648. In addition, 5-gallon pails and 55-gallon drums shall have a BLACK band approximately one-inch wide painted on the side directly below the top head chime. Exterior tops an bottoms of cans shall not be painted.

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5.8.3 Synthetic Lubricating Oils.

a. MIL L 7808, Lubricating Oil, Aircraft Turbine Engines, Base. Exterior tops, bodies and bottoms of 8-ounce, 1-quart and 1-gallon containers shall not be painted.

b. MIL-C-8188. Corrosion Preventive, Aircraft Engine, Turbine Exterior tops, bodies and bottoms of 1-quart and 1-gallon containers shall not be painted.

c. MIL-L-23699. Lubricating Oil, Aircraft and Turboprop and Turboshaft Engines, Synthetic Base. Exterior tops and bottoms of 8-ounce, 1-quart, and 1-gallon cans shall not be painted. The bodies shall be GREEN approximating No. 14110 of Federal Standard 595, and the markings shall be YELLOW approximating No. 13655.

d. MIL-L-85734, Lubricating Oil, Helicopter Transmission Systems, Synthetic Base. Exterior tops and bottoms of 1 quart and 1 gallon cans shall not be painted. The bodies shall be PURPLE approximating No. 17142 of Federal Standard 595 and the markings shall be WHITE approximating No. 17875.

5.8.4 Petroleum Lubricating Oils.

a. MIL-L-6081. Lubricating Oil, Jet Engine. Exterior tops and bottoms of 1-quart and 1-gallon hermetically sealed containers shall not be painted.

5.8.5 Brake Fluids.

a. VV-B-680. Brake Fluid, Automotive. Exterior tops and bottoms of 1-pint and 1-gallon containers shall not be painted.

b. MIL-B-46176. Brake Fluid, Silicon, Automotive All Weather, Operational and Preservative. Exterior tops and bottoms of 1-pint and 1-gallon containers shall not be painted.

5.8.6 Painting. The package (1-quart and 1-gallon metal cans) for any product having stringent particulate contamination limits shall not have coating (paint) applied to the ends of the container or within ½ inch of the ends. Side striping not required.

5.9 Marking.

5.9.1 Material and Methods. Marking shall be clear, legible, durable and nonfading. Marking of unit packs, intermediate packs and exterior packs shall be accomplished by use of labels, stamping, stenciling, printing, lithographing, silk-screening, photo marking, embossing or other similar processes.

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5.9.1.1 Labels. Labels may be used for address markings (See 5.9.4.6.19). They are permitted for contract data, and will be used when required by regulation or statute. Labels are permitted for marking identification data on unit containers, unit packs, intermediate packs and on exterior packs when the type or size of exterior pack does not permit stenciling. Printed labels are authorized for identification markings on exterior containers only when specified. When labels are authorized for use on cans, pails, or drums they must be waterproofed, regardless of level of pack cited.

5.9.2 Color of Markings. For levels B and C, markings on TAN coated containers will be BLACK. Markings on uncoated containers, such as fiberboard or plastic, and on containers coated other than TAN, shall be of contrasting color.

5.9.3 Abbreviations. The following abbreviations are authorized for use. In addition, abbreviations of the item descriptions will be permitted when approved by the procuring agency. Periods will not be used with abbreviations.

(a) Package units:

Bottle	BT
Box	BX
Cake	CK
Can	CN
Carton	CT
Container	CO
Drum	DR
Jar	JR
Pack	PK
Pail	PL
Tube	TU
Unit	UN

(b) Quantitative units:

Ampoule	AM
Dozen	DZ
Each	EA
Great Gross	GG
Gross	GR
Group	GP
Hogshead	HH
Hundred	HD
Lot	LO

(c) Weights and measure units:

Cube	CU
Cubic Centimeter	CC
Dram	DM
Fluid Ounce	FL OZ
Foot	FT
Gallon	GL
Gallon Imperial	GB
Gill	GI
Gram	G
Kilogram	KG
Liter	L
Ounce	OZ
Pint	PT
Pound	LB
Quart	QT
Quart Imperial	QI
Volume	VO
Weight	WT

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(d) Product markings:

aircraft----------ACFT
and----------&
automotive----------AUTO
engine----------ENG
gyroscope----------GYRO
lubricating----------LUBE
petroleum----------PETRO
Preservative----------PRESERV
Qualification number---QUAL
reciprocrating---------RECIP
temperature----------TEMP
turbine----------TURB

(e) Miscellaneous:

Bill of Lading----------B/L
Contract----------CONTR
Copy----------CY
Federal Supply Class----------FSC
Federal Supply Code for Manufacturers----------FSCM
Government Bill of Lading----------GBL
Dimensions----------DMN
Identification----------IDENT
Invoice----------INV
Manufactures part number----------MFR/PN
Net weight----------NET WT
Packed----------PKD
Port transportation officer----------PTO
Quantity----------QTY
Tare weight----------T/WT
Transportation Officer-TO
Less than carload------LCL
Less than truckload----LTL
Manufacture----------MFR
Mark----------MK
Month----------MO
National item identification number--------NIIN
National Stock Number--NSN

5.9.4 Identification, contract data and address markings.

5.9.4.1 Unit containers one gallon size and smaller. Identification and contract data shall be marked on unit containers one gallon size and less as specified in Table IV. (see Figure 1).

5.9.4.2 Unit and intermediate containers. Identification and contract data shall be marked on unit and intermediate containers (folding, setup and fiberboard boxes) as specified in Table IVa. (see Figure 2).

5.9.4.3 Drums and pails. Identification, contract data and address markings shall be marked on drums and pails as specified in Table IVb. (see Figure 3 and 4). When shipments are palletized, address markings are not required on each individual container. (See 5.9.4.5).

5.9.4.4 Fiberboard shipping containers. Identification, contract data and address markings shall be marked on exterior fiberboard boxes as specified in Table IVc. (see Figure 5). When shipments are palletized, address markings are not required on each individual container (see 5.9.4.5).

5.9.4.5 Unti loads. Individual shipping containers composing a unit load shall be marked with the identification markings and contract data required. All pallet loads will be marked in accordance with MIL-STD-129. (See Figure 6).

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5.9.4.6 Location and sequence of marking elements. The sequence of marking elements are shown in Figures and Tables are mandatory, but the location of the data may be varied as required by size of marking and container size and shape. Only the applicable information shall be marked on the containers, not the words in the Tables, such as "Code" or "Military Symbol". Containers not specified in this standard shall be marked to conform to the Tables as closely as practicable. Markings on containers of 5-gallon capacity or less shall occupy a minimum of 60% of the available vertical space on the sides to be marked.

5.9.4.6.1 NATO Code. The NATO Code number is an identifying designation consisting of a letter and a number assigned to a petroleum product standardized for use by the NATO Armed Forces. The NATO Code number shall be placed above the Military Symbol and enclosed ina rectangle which may be either a continuous or broken line.

5.9.4.6.2 Military Symbol. The Military Symbol shall be the identifying designation assigned in a particular product specification.

5.9.4.6.3 Item description. The item description shall be the exact name and description of the item as it appears in the contract.

5.9.4.6.4 Specification number. The specification number shall be that shown by the contract or order and shall include the revision and amendment designation if any. Nonspecification products such as "brand name or equal" shall be identified with the appropriate brand name and part number as cited in the contract.

5.9.4.6.5 Type, class and grade. The applicable type, class and grade of product as designated in the product specification shall be shown. Type, class and grade may be omitted from markings when they are the same as the Military Symbol.

5.9.4.6.6 National Stock Number. The stock number shall be the exact National Stock Number (NSN) consisting of 13 digits (9150-00-000-0000). When no NSN is available, the applicable Federal Supply Classification (FSC), if known, shall be shown. In addition, when no NSN is available, the manufacturer's part number (MFR/N) shall be used and space shall be left blank immediately above the number for subsequent placement of NSN. When the MRF/PN is used, the Federal Supply Code for Manufacturers (FSCM) appropriate to the part number shall precede the part number on the same line. The letters FSCM shall be used to identify the code indicated. (The FSCM is cross-referenced to manufacturer's names and address in Cataloging Handbooks H4-1 and H4-2).

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5.9.4.6.7 Shelf-life markings. There are two types of shelf-life items (see 3.6) and when specified in the contract, the type of shelf-life item shall be marked below the item identification data on exterior packs (fiberboard shipping boxes, drums and pails). (See 6.1)

a. Marking for Type I shelf-life items shall be as follows:

Date of Manufacture _____
Expiration Date _____

Instructions:

For the Date of Manufacture, show the actual date, month and year.

For the Expiration Date, show the month of manufacture plus the number of months shelf-life assigned in contract or order.

b. Marking for Type II shelf-life items shall be as follows:

Date of Manufacture _____
Test Date _____
New Test Date _____

Instructions:

For the Date of Manufacture, show the date of manufacture or the middile month of quarter in which material is manufactured.

For the Test Date, show the date of manufacture plus the number of months shelf-life assigned in the contract order.

After the New Test Date, leave a blank.

5.9.4.6.8 Date of Manufacture. When shelf-life marking is not required the date of manufacture of contents shall be designated by the center month of each quarter and year (namely, February 1985, May 1985, August 1985, and November 1985) in which most of the manufacturing is done, or the actual month and year manufacturing took place (e.g., January 1984, April 1984, December 1984). The date of manufacture may be indicated on the lids of Types I through V cans by embossing or other marking methods.

5.9.4.6.9 Batch or lot number. The batch or lot number shall be that series of numbers or letters or both established by the manufacturer to record production and control of product. Where both lot and batch numbers are applicable, both shall be shown. The word "BATCH" or "LOT" shall precede the applicable batch or lot number on all containers of all products. In those instances where the batch or lot number is manually stenciled on the side of the container, it will be applied in such a manner, utilizing good quality material, as to preclude any tendency to obliterate or smear. Batch or lot number may be indicated on the lids of Type I through V cans by embossing or other marking methods.

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5.9.4.6.22 Contractor's name. Where the contractor's name requires two or more lines, a readily identifiable abbreviation is permitted.

5.9.4.6.23 Bar coding. All units, intermediate and shipping containers will be marked with bar coding symbols in accordance with MIL-STD-1189, and located in accordance with MIL-STD-129.

5.10 Returnable Drum Deposit Marking. Drums returnable to the supplier for a deposit refund shall be conspicuously marked to indicate the amount of deposit and the location to which it is returnable.

5.11 Hazardous materials data. When specified in the contract, a Material Safety Data Sheet shall be prepared and submitted in accordance with Federal Standard No. 313. (See 6.1).

6. NOTES

6.1 Ordering Data

4.3 State Level B or C, or Commercial packing.

4.5 Specify palletization, if required.

4.6 Specify when Military Air Shipment requirements also apply.

5.1.1 State Level B or C, or commercial unit packing.

5.2.1 State Level B or C, or Commercial unit packing.

5.2.3 State Level B or C, or Commercial packing.

5.3.1 State Level B or C, or Commercial packing.

5.4 State Level B or C, or Military Airlift Requirements. When Military Airlift is required the contracting officer shall provide necessary instructions and forms in accordance with Air Force Regulation 71-4.

5.4.1 Specify palletization, if required.

5.5 State Level B or C, or Commercial packing.

5.6.1 State Level B or C, or Commercial packing.

5.6.3 State Level B or C, or Commercial packing.

5.7.1.2 State Level B or C, or Commercial exterior packing.

5.7.2.2 State Level B or C, or Commercial exterior packing.

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5.9.4.6.12 Net Contents. Show the net contents, by volume or by weight in pounds, as appropriate to the commodity.

5.9.4.6.17 Specify units of issue to the unit pack, intermediate pack and shipping container.

5.9.4.6.18 Specify the unit of issue, unit of measure and quantity to be used.

5.9.4.6.19 Specify address markings to be used.

5.11 When a Material Safety Data Sheet is required, include DAR clause and specify the mailing address.

Custodians
Army - SM
Navy - SA
Air Force - 68

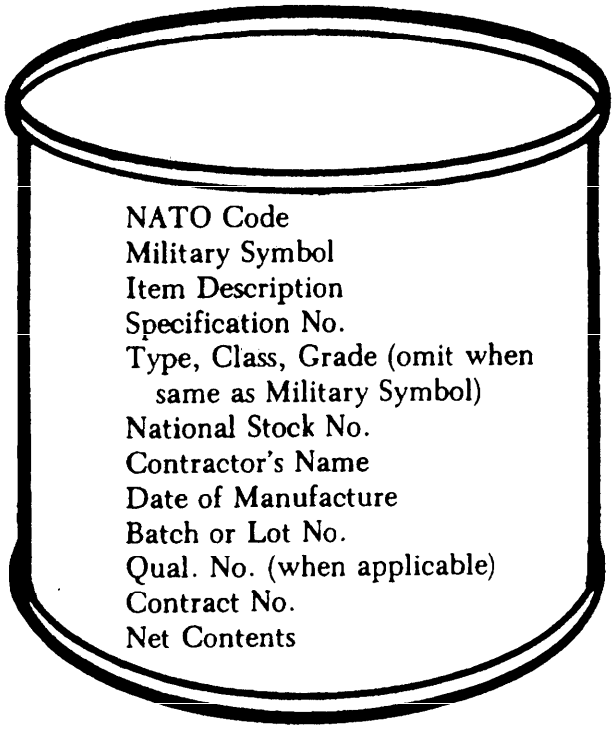
Review Activities
Army - MI, MR,
Navy - AS, SH, OS, YD, MC,
DLA-PA

Preparing Activity
Navy - SA
Project No.
PACK 0570

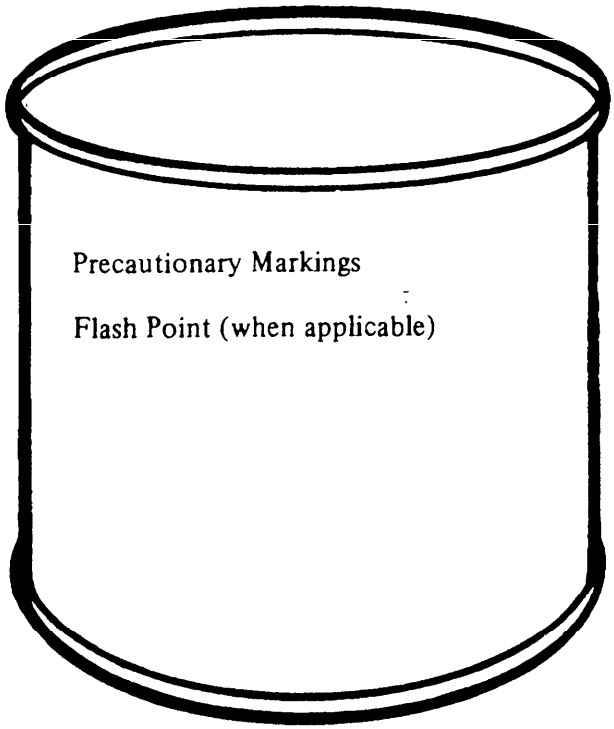
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The sequence and location of marking elements as shown in this Figure are desirable but the sequence of data and location may be varied as required by size of markings and container size and shape.



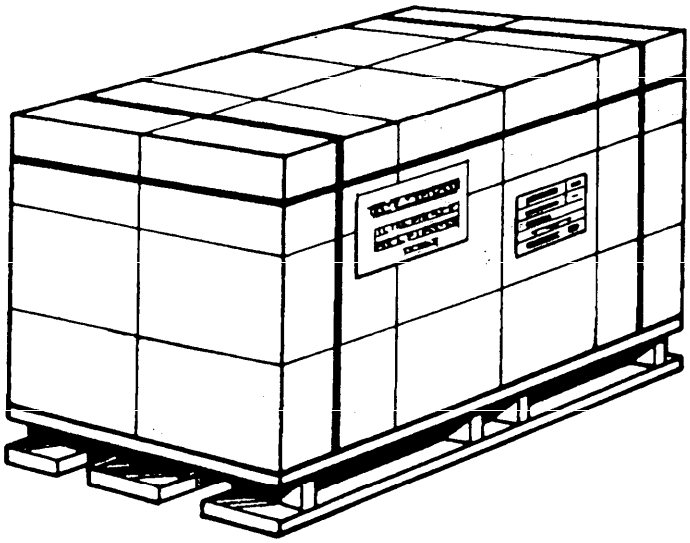
Side

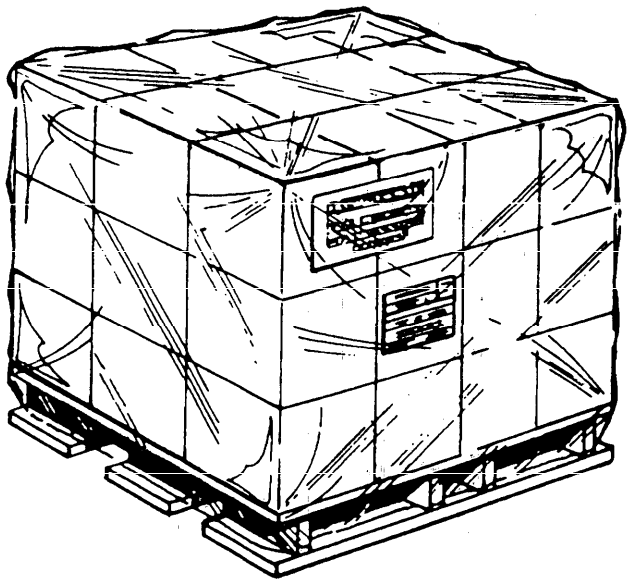


Opposite Side

FIGURE 1 – Marking Unit Containers (one gallon and less) for Level B Shipments.

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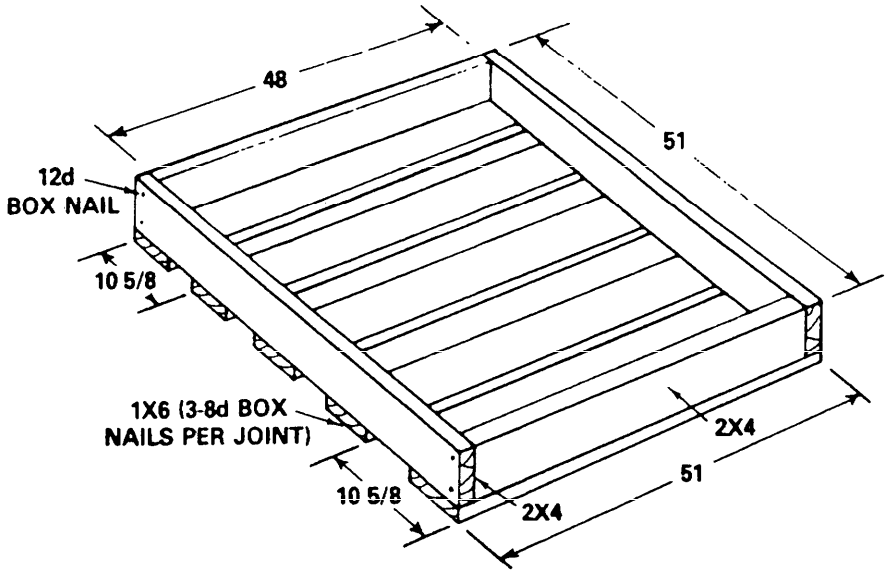




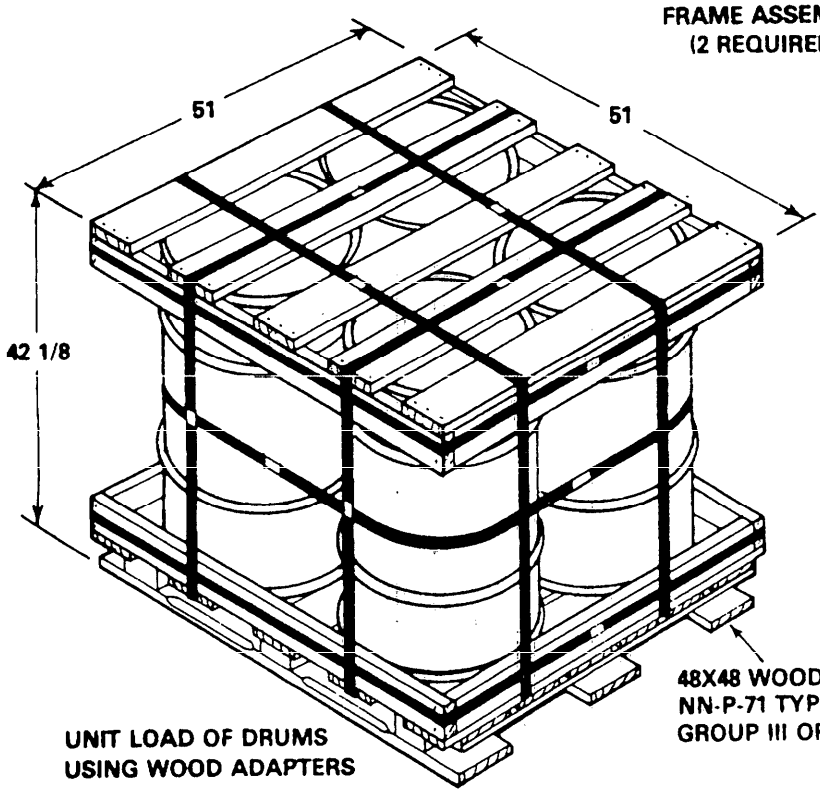
For pallet marking see paragraph 5.9.4.5

FIGURE 6 – Marking Palletized Shipments of Petroleum and Related Products

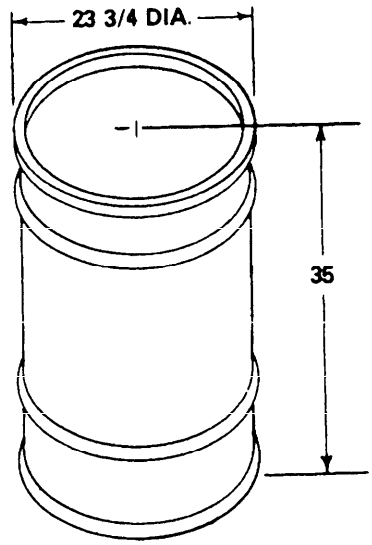
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FRAME ASSEMBLY
(2 REQUIRED)



UNIT LOAD OF DRUMS
USING WOOD ADAPTERS



DRUM

NOTES:

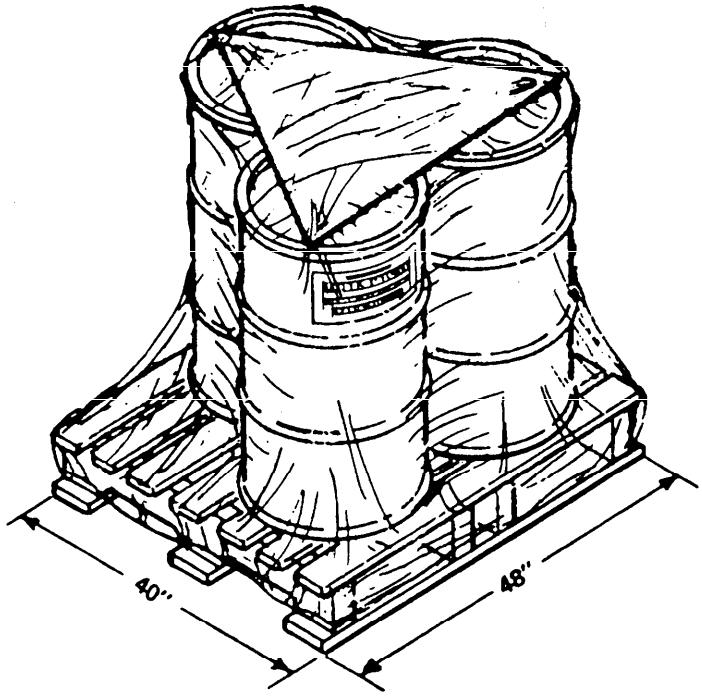
1. TIE-DOWN STRAPS. Unit Loads Shall Have a Minimum of Four Tie-Down Straps with Two Straps Running the Length of the Unit Load and Two Straps Running the Width of the Unit Load.

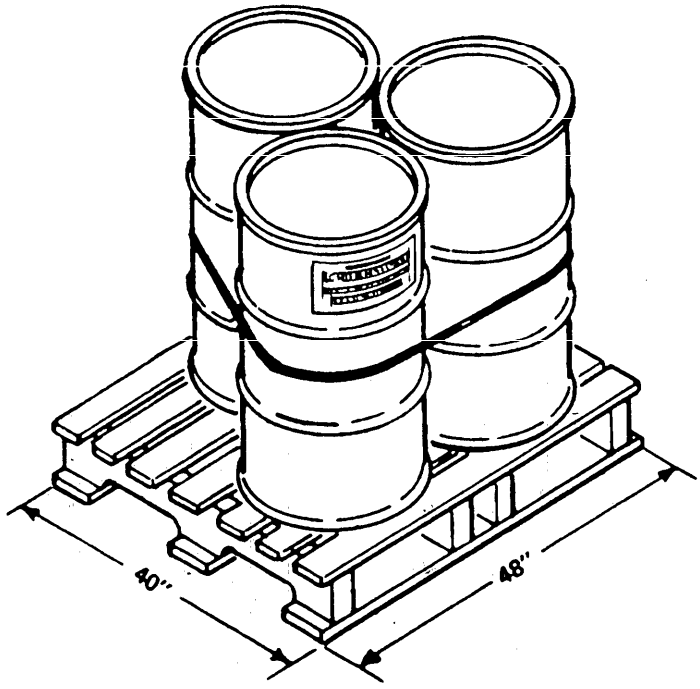
2. STEEL STRAPPING. 3/4 Inch Strapping (QQ-S-781) Shall Be Used.

55 GALLON DRUM PALLETIZATION

FIGURE 7

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ARMY USE ONLY
55 Gallon Drum Palletization

FIGURE 7A

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TABLE I – UNIT CONTAINERS FOR PETROLEUM AND RELATED PRODUCTS

UNIT CONTAINER	SPECIFICATION REQUIREMENTS		COMMERCIAL	CLOSURE AND SEALS	COATING REQUIREMENTS
	LEVEL B	LEVEL C			
Bottles	Bottles shall be round, oblong or rounded square with sloping shoulders. Side walls shall be somewhat thicker than the shoulders. Bottle openings shall be of sufficient size to permit easy removal of contents. PPP-C-186	Same as for level B	Bottles normally used for the product being furnished	Adequate to prevent leakage	Not Applicable
Cans, Composite with metal ends		PPP-C-1581, Class II, Type I	ANSI MH 3.4		Not Applicable
Cans, metal 28 gage and lighter	PPP-C-96 Type I – Round, open top, double seamed ends Type III – Open-top double seamed ends key opening with reclosure feature Type IV – Round, square, or oblong, both ends double seamed on Type V, Class 2 – multiple friction plug Type V, Class 4 – screw cap Type V, Class 9 – Press-in closure with plastic push-pull spout Type VIII – Round, dome or cone top, both ends double seamed onto body Type IX – Pressurized Type IX, Class 1-Round can, soldered side seam, double-seamed concave top and bottom Type IX, Class 2-Round can, soldered side seam, double-seamed concave dome top with 1-inch opening Type IX, Class 3-Round can, draw seamless body & dome top with 1-in. opening & double seamed bottom Metallic Coating weight of cans shall be of minimum 1.25 pound hot dipped tinplate or minimum 0.25 pound electrolytic tinplate.	PPP-C-96 same as for Level B	ANSI 3.1 for Round Metal Motor Oil Cans	Inner seals shall be furnished with closure (Type V, Class 4 of PPP-C-96) for containers of one-pint capacity or greater. Type IX cans shall be fitted with a valve designed for dispensing a surface spray. Class 2 and 3 cans shall be fitted with a cap to protect the valve. Cans containing suspending solids which may settle out shall contain a metal ball which will facilitate agitating the solids into suspension.	Exterior coating of cans (tops, bodies and bottoms) and metal screw caps shall be in accordance with the color and coating requirements in 5.9.

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TABLE I – UNIT CONTAINERS FOR PETROLEUM AND RELATED PRODUCTS

UNIT CONTAINER	SPECIFICATION REQUIREMENTS		COMMERCIAL	CLOSURE AND SEALS	COATING REQUIREMENTS
	LEVEL B	LEVEL C			
	Side seams may be cemented or soldered Type 1-quart and 1-gallon cans may also have welded side seams. Type I cans shall be fabricated from minimum 35 gage (MSC) tinplate Type V, Class 4 and Class 8 or 9 cans of 1-gallon capacity shall be provided with a handle affixed to the top of the can. Cans shall be of commercial sizes as regularly supplied by the petroleum industry for the product being furnished.				
Cartridges, Grease	MIL-G-22587 except that cartridges do not have to be metal Both ends of the cartridges shall be unpainted Each cartridge shall be marked just below the removable cover "Remove cover and place this end in gun first"	MIL-G-22587	ANSI MH 3.3		Not applicable
Drums, steel (120 pounds grease), full-removable lug cover head	PPP-D-705, 20 Gauge, Type III	PPP-D-705, 22 Gauge, Type III	ANSI MH 2.8 1979	Adequate to prevent leakage. Gasket required.	The exterior coating of drums shall be in accordance with the color and coating requirements specified in 5.9. Side seams may be coated with a clear lacquer.

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TABLE I – UNIT CONTAINERS FOR PETROLEUM AND RELATED PRODUCTS

UNIT CONTAINER	SPECIFICATION REQUIREMENTS		COMMERCIAL	CLOSURE AND SEALS	COATING REQUIREMENTS
	LEVEL B	LEVEL C			
Drums, steel, 16 gallon full-removable head	PPP-D-705, Type VI	Same as for Level B	ANSI MH 2.14 1979	Adequate to prevent leakage. Gasket required.	The exterior coating of drums shall be in accordance with the color and coating requirements specified in 5.9. Side seams may be coated with a clear lacquer
Drums, steel 55 gallon closed head	PPP-D-729, 16 Gauge, Class A	PPP-D-729, 18/18 Gauge Type II, Class A	ANSI MH 2.1 1979	Adequate to prevent leakage. Drums used for MIL-L-17331 and MIL-H 17672 shall not be fitted with zinc coated closures.	For contractor-furnished drums, fog spraying of interior surfaces is not mandatory. Drums intended for filling with aviation fuels shall not have interior surfaces coated with any materials that are soluble in aviation fuels. If these drums are fog-sprayed, they must be steamed cleaned and dried before filling. Drums used for MIL-L-17331 and MIL-H-17672 shall not have interiors coated with zinc phosphate. The exterior coating of drums shall be in accordance with the color and coating requirement specified in 5.9.

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TABLE I – UNIT CONTAINERS FOR PETROLEUM AND RELATED PRODUCTS

UNIT CONTAINER	SPECIFICATION REQUIREMENTS		COMMERCIAL	CLOSURE AND SEALS	COATING REQUIREMENTS
	LEVEL B	LEVEL C			
Drums, steel 55 gallon, open head	PPP-D-729, Type III/IV as specified	Same as for Level B	ANSI MH 2.5 1979	Adequate to prevent leakage. Gasket required.	For contractor-furnished drums, fog spraying of interior surfaces is not mandatory. The exterior coating of drums shall be in accordance with color and coating requirements specified in 5.9.
Jars	Jars shall be round, squat type, with opening wide enough to permit easy removal of all contents with a spatula	Same as for level B	Jars normally used for the product being furnished	Adequate to prevent leakage.	Not Applicable
Pails, metal fully removable cover	PPP-P-704, Type II, Class 8	PPP-P-704, Type II, Class 3	ANSI MH 2.15 1979	Adequate to prevent leakage. Gasket required.	The exterior coating of pails shall be in accordance with the color and coating requirements specified in 5.9.
Pails, metal tight head	PPP-P-704, Type I, Class 3, 24 gage, 1-5 gallon. Class 8 22 gage, 10 gallon. Wire handles shall be zinc coated or otherwise protectively coated to resist corrosion.	Same as for Level B	ANSI MH 2.9 1979	A vented closure, illustrated in figures 9, 10, 11 or 12 of PPP-P-704, or a 1-3/4" screw-cap closure equipped with an inner seal and a separate 1/2" vent tube installed directly opposite the closure shall be used. The vent will be installed by the pail manufacturer and equipped with a cap attached to the vent tube. Only metal self-venting closures are acceptable for aviation engine lubricating oils and aviation hydraulic fluids. Self-venting closures for products con-	The exterior coating of pails shall be in accordance with the color and coating requirements specified in 5.9.

TABLE I – UNIT CONTAINERS FOR PETROLEUM AND RELATED PRODUCTS

UNIT CONTAINER	SPECIFICATION REQUIREMENTS		COMMERCIAL	CLOSURES AND SEALS	COATING REQUIREMENTS
	LEVEL B	LEVEL C			
				forming to the following specifications shall conform to figures 9 or 10 of PPP-P-704 and need not be colored olive drab: MIL-L-2104, MIL-L-2105, MIL-L-9000, MIL-L-14107, MIL-L-21260, MIL-L-46152.	
Tubes, collapsible	PPP-T-1637, Type I – Metal, Type II – Plastic, or Type III – Laminated polyethylene-aluminum foil-paper	Same as for level A	Collapsible tubes normally used for the product being furnished	The caps shall be made from thermo-set, thermo-plastic or the same material as used for the tube.	Not Applicable

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TABLE II Classification of Cans and Requirements for Arrangement in Interior and Exterior Packs

Size of Cans	Number of Intermediate Packs	Arrangement of Intermediate Packs in Exterior Packs		Arrangement of Cans in Intermediate or Exterior pack		Total Number of Cans
		Floored	Layers	Floored	Layers	
Composite and Type I and VIII, round Cans						
8-ounce	2	1x2	1	4x3	3	72
1-quart	None	–	–	4x3	2	24
1-gallon	None	–	–	2x3	1	6
Type III, open-top, key-opening with reclosure feature cans						
1-pound	None	–	–	4x3	2	24
Type V, Class 2 and VIII, round cans						
4-ounce	4	1x2	2	3x2	4	96
8-ounce	6	1x2	3	3x4	1	72
1-pound	None	–	–	3x4	2	24
5-pound	None	–	–	3x2	1	6
1-pint	None	–	–	4x6	2	48
1-quart	None	–	–	4x3	2	24
1-gallon	None	–	–	2x3	1	6
Type V, Class 4, 6, 8 and 9 and VIII round and oblong cans						
2-ounce	4	1x2	2	3x8	2	192
4-ounce	4	1x2	2	4x6	1	96
8-ounce	6	1x3	2	2x6	1	72
1-pint	None	–	–	12x4	1	48
1-quart	None	–	–	4x6	1	24
1-gallon	None	–	–	2x3	1	6
Type IX, Class 1, 2, and 3. Cans, pressurized with valve						
12-ounce	2	1x2	1	4x6	1	48
16-ounce	2	1x2	1	4x6	1	48

TABLE III Packing Arrangement of Tubes, Bottles and Jars

Intermediate pack arrangement					Exterior pack arrangement				
Unit weight of contents	No. of unit containers	Arrangements of unit containers			No. of inter-mediate packs	Arrangement of packs			No. of unit containers
		No. of layers	No. of rows	No. in row		No. of layers	No. of rows	No. in row	
2-ounces or less. . . .	48	2	4	6	4	1	2	2	192
4-ounces	24	1	4	6	4	2	1	2	96
6-ounces	24	1	4	6	4	2	1	2	96
8-ounces	24	1	4	6	2	2	1	1	48
1-pound.	NA	NA	NA	NA	NA	1	4	6	24
1-quart	NA	NA	NA	NA	NA	1	3	4	12

[1] If the unit quantities specified in the contract or order are by volume, they shall be converted to their corresponding weights for applying this table. Where the weight is not listed, the next higher weight given in the table shall be used.

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TABLE IV. IDENTIFICATION AND CONTRACT DATA MARKING ON UNIT CONTAINERS, ONE GALLON AND LESS

UNIT CONTAINERS (ONE GALLON AND LESS)	MARKING ELEMENTS				
	LEVEL B	REFERENCE	LEVEL C	REFERENCE	COMMERCIAL
5-pound can 1-gallon can 1-pound can 1-quart can 1-pint can 16-ounce can 12-ounce can 8-ounce can 4-ounce can 2-ounce can bottles jars collapsible tubes grease cartridges (see Note 1)	NATO Code (when assigned)	5.9.4.6.1	NATO Code (when assigned)	5.9.4.6.1	Markings offered by the industry will be acceptable on unit containers providing they conform to ASTM D 3951-82
	Military Symbol (when assigned)	5.9.4.6.2	Military Symbol (when assigned)	5.9.4.6.2	
	Item Description	5.9.4.6.3	Item Description	5.9.4.6.3	
	Specification Number	5.9.4.6.4	Specification Number	5.9.4.6.4	
	Type, Class, Grade (omit when same as Military Symbol)	5.9.4.6.5	Type, Class, Grade (omit when same as Military Symbol)	5.9.4.6.5	
	National Stock Number	5.9.4.6.6	National Stock Number	5.9.4.6.6	
	Contractor's Name	5.9.4.6.22	Contractor's Name	5.9.4.6.22	
	Date of Manufacture (see Note 2)	5.9.4.6.8	Date of Manufacture (see Note 2)	5.9.4.6.8	
	Batch or Lot Number (see Note 2)	5.9.4.6.9	Batch or Lot Number (see Note 2)	5.9.4.6.9	
	Contract Number	5.9.4.6.10	Contract Number	5.9.4.6.10	
	Qualification Number (see Note 2)	5.9.4.6.11	Qualification Number (see Note 2)	5.9.4.6.11	
	Net Weight	5.9.4.6.12	Net Weight	5.9.4.6.12	
	Precautionary Markings (when req.)	5.9.4.6.16	Precautionary Markings (when req.)	5.9.4.6.16	
	Flash Point (when required)	5.9.4.6.15	Flash Point (when required)	5.9.4.6.15	
	Bar Code	5.9.4.6.23	Bar Code	5.9.4.6.23	

NOTES: 1. Markings on cartridges shall be applied either vertically or horizontally. Each cartridge shall have an instruction just below the removable cover reading "Remove cover and place this end in gun first".

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TABLE IVa. IDENTIFICATION AND CONTRACT DATA MARKINGS ON UNIT AND INTERMEDIATE CONTAINERS (FOLDING, SETUP, AND FIBERBOARD BOXES)

MARKING ELEMENTS (LEVELS B AND C AND COMMERCIAL)	REFERENCE	SIZE
NATO Code (when assigned)	5.9.4.6.1	Legible
Military Symbol (when assigned)	5.9.4.6.2	Legible
Item Description	5.9.4.6.3	Legible
Specification Number	5.9.4.6.4	Legible
Type, grade, class (omit when same as Military Symbol)	5.9.4.6.5	Legible
National Stock Number	5.9.4.6.6	Legible
Quantity	5.9.4.6.17	Legible
Unit of Issue	5.9.4.6.18	Legible
Precautionary Marking (when required)	5.9.4.6.16	Legible
Flash Point (when required)	5.9.4.6.15	Legible
Bar Code	5.9.4.6.23	Legible
Date of Manufacture	5.9.4.6.8	Legible
Batch or Lot Number	5.9.4.6.9	Legible

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TABLE IVb. MARKINGS ON DRUMS AND PAILS: LEVELS B AND C, AND COMMERCIAL

MARKING ELEMENTS	SIZE	REFERENCE	SIDE MARKINGS			TOP MARKINGS		
			DRUMS		PAILS	DRUMS		PAILS
			CLOSED HEAD	OPEN HEAD (SEE NOTE 2)		CLOSED HEAD	OPEN HEAD	
NATO (when assigned)	3/4	5.9.4.6.1		X	X	X	X	
Military Symbol (when assigned)	1	5.9.4.6.2	X	X	X	X	X	
Item Description	1/2	5.9.4.6.3	X	X	X	X	X	
Specification Number	1/2	5.9.4.6.4	X	X	X	X	X	
Type, Class, Grade (omit when same as Military Symbol)	1/2	5.9.4.6.5	X	X	X	X	X	
National Stock Number	1/2	5.9.4.6.6		X	X	X	X	
Overseas Address (see Note 1)	1	5.9.4.6.19			Opposite side	X	X	
Domestic Address	1	5.9.4.6.19	X	X				X
Date of Manufacture	Legible	5.9.4.6.8		X	X	X	X	
Contractors Name	Legible	5.9.4.6.22	X	X	X			
Batch or Lot Number	Legible	5.9.4.6.9		X	X	X	X	
Contract Number	Legible	5.9.4.6.10		X	X	X	X	
Qualification Number	Legible	5.9.4.6.11		X	X	X	X	
Net Weight	Legible	5.9.4.6.12		X	X	X	X	
Gross Weight	Legible	5.9.4.6.20		X	X	X	X	
Cube	Leigible	5.9.4.6.21		X	X	X	X	
Shelf Life	1/2	5.9.4.6.7	X	X	Opposite side			
Precautionary Marking (when required)	1	5.9.4.6.16	X	X	Opposite side			
Flash Point (when required)	1/2	5.9.4.6.15	X	X	Opposite side			
Regulatory Markings	As required by regulation	5.9.4.6.13	X	X	Opposite side			

AN X INDICATES MARKINGS ARE REQUIRED IN THE LOCATION SHOWN

NOTE: 1. Overseas address shall be placed on side of drum opposite other side of markings if top head construction will not permit such marking.
2. When space does not permit placing required markings on the 400-pound drum between the rolling hoops, the NATO symbol and the Military symbol, when assigned, may be placed immediately above the upper rolling hoop.

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TABLE IVc. IDENTIFICATION, CONTRACT DATA AND ADDRESS MARKINGS ON EXTERIOR FIBERBOARD BOXES FOR PACKED PETROLEUM AND RELATED ITEMS (LEVELS B AND C AND COMMERCIAL)

MARKING AND LOCATION	SIZE OF LETTERS (INCHES)	REFERENCE
SIDE (left upper two-thirds)		
NATO Code		
Military Symbol (when assigned)	1/2	5.9.4.6.1
National Stock Number	1/2	5.9.4.6.2
Item Description	1/2	5.9.4.6.6
Specification Number	1/2	5.9.4.6.3
Type, grade, class (omit when same as Military Symbol)	1/2	5.9.4.6.4
Quantity	1/2	5.9.4.6.5
Unit of Issue	Legible	5.9.4.6.17
Gross weight and cube	Legible	5.9.4.6.18
Date of Manufacture (not required when shelf life is specified)	1/2	5.9.4.6.20/21
Batch or Lot Number	1/2	5.9.4.6.8
Shelf-life (when specified) (see Note 1)	1/2	5.9.4.6.9
	1/2	5.9.4.6.7
SIDE: (below identification marking)		
Contract Number		
Qualification Number (when applicable)	1/2	5.9.4.6.10
Contractor's Name	1/2	5.9.4.6.11
	1/2	5.9.4.6.22
SIDE (right lower two-thirds)		
Address Marking (see Note 2)		
Flash Point (when required)	1/2	5.9.4.6.19
Regulatory Markings (When required)	1/2	5.9.4.6.15
	as required by CFR 49	5.9.4.6.13

NOTES: 1. Shelf life markings may be marked on end when space does not permit marking on side.
2. Address markings may be marked on opposite side when space does not permit marking on identification marked side. On palletized loads, the address marking need not be marked on each individual container. (See 5.9.4)